



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12

1998 APRIL SCHOOL EXAMINATIONS

MATHEMATICS

2/3 UNIT (COMMON)

Time Allowed - Three Hours
(Plus 5 minutes reading time)

DIRECTIONS TO STUDENTS:

1. Attempt ALL questions.
2. Show all necessary working.
3. Begin each question on a new sheet of paper.
4. Place your name, class and teacher at the top of each page.
5. Mark values are shown at the side of each question.
6. Non-programmable calculators are permitted.
7. Standard integrals are printed on the last page.

Question 1.	Start each question on a new page.	Marks
a) (i)	Write the basic numeral for 4.17×10^{-3} .	1
(ii)	Express $\sqrt{0.003615}$ correct to 3 significant figures.	1
b)	Simplify $\sqrt{12} + \sqrt{27} - \sqrt{3}$	2
c)	Solve: $x + 2y = 1$ $3x + y = 13$	2
d)	Find the exact value of $\sin 120^\circ + \cos 300^\circ$	2
e)	Solve: $ x - 4 > 2$	2
f)	The retail price of a T.V. is increased by 15% to \$1104. What was the price before the increase?	2

Question 2.	Start each question on a new page.	Marks
a)	Simplify $(x - y)^2 - (x + y)^2$	2
b)	Express $\frac{2}{3 + \sqrt{7}}$ with a rational denominator and simplify.	2
c)	Factorise completely $8y^3 + 8$.	2
d)	If $v^2 = u^2 - 2as$, find the value of s if $v = 3.26$, $u = 1.73$ and $a = -4.6$ (Give your answer correct to 2 decimal places.)	2
e)	Find the value of $\lim_{m \rightarrow -2} \frac{m + 2}{m^2 - 4}$	2
f)	Solve: $x + \frac{4 - x}{3} = 12$	2

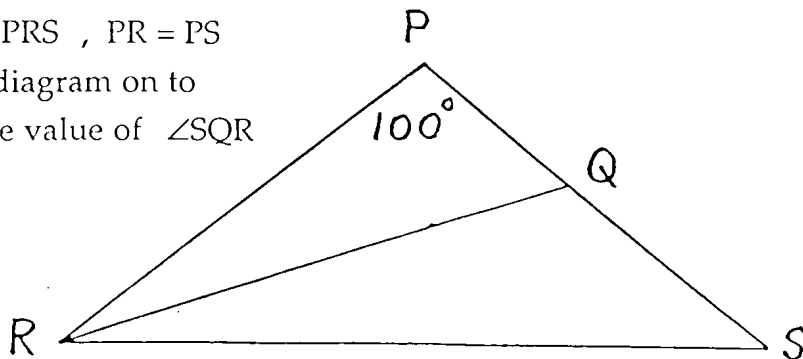
Marks

2

- Find the length of AB and the gradient of AB . 2
- Show that the equation of the line l , drawn through C parallel to AB is $2x + y + 4 = 0$. 2
- Find the coordinates of D , the point where the line l intersects the x -axis. 2
- M is the midpoint of AC . Find the coordinates of the point M . 1
- Find the distance of the point A from the line l . 2
- Find the area of the quadrilateral $ABCD$. 3

Marks

- a) Show that the expression $2x^2 - 4x + 5$ is positive definite. 2
- b) If α and β are the roots of the equation $3x^2 + 2x - 6 = 0$, find the value of:
- (i) $\alpha^2\beta^2$ ii) $\alpha^2 + \beta^2$ 3
- c) On a number plane shade the region satisfied simultaneously by $x + 2y \geq 2$ and $y \leq 4 - x^2$. 3
- d) In the diagram QR bisects $\angle PRS$, $PR = PS$ and $\angle RPS = 100^\circ$. Copy the diagram on to your exam paper and find the value of $\angle SQR$
-
- 2



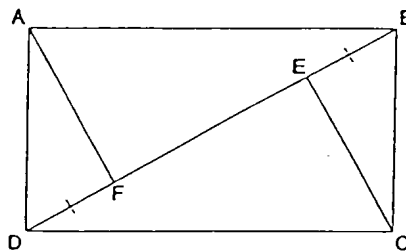
- e) Evaluate $\sum_{k=1}^{30} (2k-1)$ 2

Question 5.	Start each question on a new page.	Marks
a) Differentiate :		
(i) $(x+3)(x^2-1)$	(ii) $\sqrt[3]{x} + 3x$	3
b) Solve $5^{2x-1} = 1$		2
c) In the triangle ABC, the length of AB is 14.1 cm; BC = 19.2 cm and $\angle BAC = 83^\circ$.		
i) Find the size of $\angle BCA$ correct to the nearest degree.		2
ii) Calculate the area of $\triangle ABC$.		2
d) Find the equation of the normal to the curve $y = \sqrt{x}$ at the point where $x = 4$.		3

Question 6.	Start each question on a new page.	Marks
a) State the domain and range of the function $y = \sqrt{x-3}$		2
b) Simplify $(1 - \sin^2 \theta)(\tan^2 \theta + 1)$		2
c) Solve the equation: $2 \sin \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.		2
d) i) On the same coordinate axes draw the graphs of $y = \sin 2x$ and $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.		2
ii) Find the number of solutions to the equation $\sin 2x = \cos x$ in the domain $0^\circ \leq x \leq 360^\circ$.		1
e) If $f(x) = 4x^3 - 3x^2 + 6x - 1$, find $f''(x)$ and hence evaluate $f''(-1) - f'(1)$.		3

Question 7. Start each question on a new page. Marks

- a) Which term of the sequence $-5, 3, 11, 19, \dots$, is 283? 2
- b) For the sequence $\frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$ find :
 i) the common ratio, 1
 ii) the limiting sum, 1
- c) Find the area enclosed by the curves $y = x + 2$ and $y = x^2$. 4
- d) ABCD is a rectangle. E and F are two points on the diagonal BD such that $BE = DF$.



Copy the diagram on to your exam paper. 4

- i) Prove that $\triangle AFD \cong \triangle CEB$ ii) Hence prove $\angle DAF = \angle BCE$.

Question 8. Start each question on a new page. Marks

- a) Find the values of k for which the equation $x^2 - (k + 3)x + (k + 6) = 0$ has equal roots. 2
- b) A battleship A is 230 km due west of a lighthouse L. It travels a distance of 110 km on a bearing of $N62^\circ E$ to a position C. Calculate the distance from the lighthouse to the battleship's position at C to the nearest kilometre. 3
- c) For the curve $y = 2x^3 - 6x^2 - 18x + 1$ 7
- Find the stationary points and determine their nature.
 - Find the coordinates of any points of inflexion.
 - For what values of x is the curve increasing?
 - Sketch the curve in the domain $-2 \leq x \leq 5$.
 - For what values of x is the curve concave down?

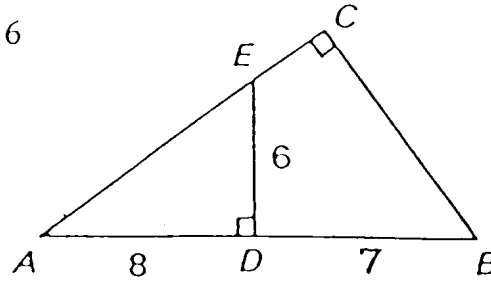
Question 9. Start each question on a new page.

Marks

a) Find: ~~7~~ i) $\int \frac{x^3 + 6}{x^2} dx$ ii) $\int_3^5 (2x + 1) dx$

4

b) $ED \perp AB$, $BC \perp AC$, $AD = 8$, $BD = 7$, $DE = 6$



i) Prove $\triangle ABC \sim \triangle AED$

2

ii) Hence or otherwise, find the lengths of BC and EC .

3

c) If $A(x - 1)(x - 2) + B(x - 1) + C \equiv 2x^2 - 3x + 5$, find A , B , and C .

3

Question 10. Start each question on a new page.

Marks

a) The gradient function of a curve is given by $\frac{dy}{dx} = 2x - \frac{4}{x^2}$. The curve passes through the point $(2, 3)$. Find the equation of the curve.

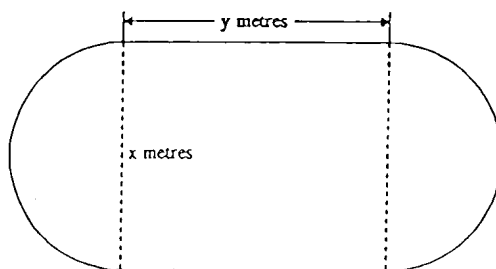
3

b) For what values of x is $x^2 \geq (x + 1)(x + 2)$.

2

c) A railway enthusiast designs a miniature railway of length 1000 metres. The route consists of two semi-circles at opposite ends of a rectangle.

7



i) If the rectangle has a length of y metres and its width is x metres, show that $y = 500 - \frac{\pi x}{2}$.

ii) Show that the area, A , enclosed by the railway track is given by $A = \frac{2000x - \pi x^2}{4}$.

iii) Find the maximum area enclosed by the railway track.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 4

A) $\Delta = b^2 - 4ac$

$$= (-4)^2 - 4(2)(5)$$

$$= 16 - 40$$

$$= -24 \quad a > 0$$

$\therefore$ positive definite (2)

B) $\alpha^2 \beta^2 = (\alpha \beta)^2$

$$= \left(\frac{c}{a}\right)^2$$

$$= \left(-\frac{6}{3}\right)^2$$

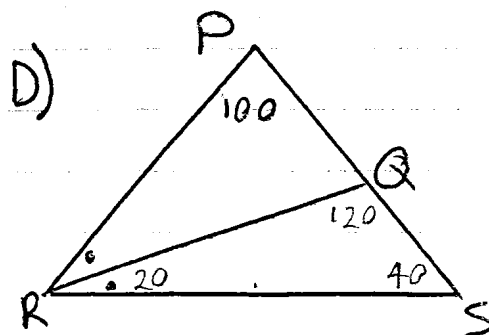
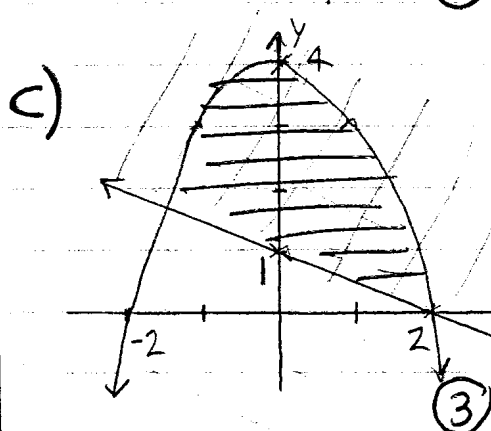
$$= 4 \quad (1)$$

I) $(\alpha^2 + \beta^2) = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(-\frac{b}{a}\right)^2 - 2x^{-2}$$

$$= \left(-\frac{2}{3}\right)^2 + 4$$

$$= 4\frac{4}{9} \quad (2)$$



$LS = LPRQ = x$ $\angle$'s opp = sides are =

$$2x + 100 = 180$$

$$x = 40^\circ$$

$\angle PRQ = \angle SRQ$ RQ bisects $\angle PRS$

$$\angle QRS = 20^\circ$$

$$\angle RQS + 20 + 40 = 180 \quad \angle \text{sum of } \triangle QRS$$

$$\angle RQS = 120^\circ \quad (2)$$

E) $1 + 3 + 5 + \dots + 59$

$$a=1 \quad d=2 \quad n=30 \quad L=59$$

$$S_n = \frac{n}{2} (a+L)$$

$$= \frac{30}{2} (1+59)$$

$$= 900$$

Question Five

AI) $y = (x+3)(x^2-1)$

$$= x^3 + 3x^2 - x - 3$$

$$y' = 3x^2 + 6x - 1$$

II) $y = x^{\frac{1}{3}} + 3x$

$$y' = \frac{1}{3} x^{-\frac{2}{3}} + 3$$

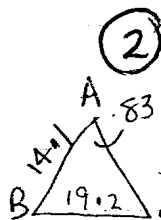
$$= \frac{1}{3\sqrt[3]{x^2}} + 3$$

B) $5^{2x-1} = 1$

$$5^{2x-1} = 5^0$$

$$2x-1 = 0$$

$$x = \frac{1}{2}$$



C) $\sin C = \sin A =$

$$\sin C = \frac{\sin 83}{14.1} = \frac{\sin 83}{19.2}$$

$$\sin C = \frac{14.1 \sin 83}{19.2}$$

$$C = 47^\circ \quad (2)$$

II) $\angle B = 180 - 83 - 47$

$$= 50^\circ$$

$$A = \frac{1}{2} ac \sin B$$

$$= \frac{1}{2} \times 14.1 \times 19.2 \times \sin 50$$

$$= 103.69 \text{ cm}^2 \quad (2)$$

D) $y = x^{\frac{1}{2}}$

$$y' = \frac{1}{2} x^{-\frac{1}{2}}$$

when $x = 4$

$$y = 2 \quad y' = \frac{1}{2} \times \frac{1}{\sqrt{4}}$$

$$m = \frac{1}{4}$$

$$y-2 = \frac{1}{4}(x-4)$$

$$4y-8 = x-4$$

$$x-4y+4=0 \quad (3)$$

Question six

A) D: $x \geq 3$

R: $y \geq 0$ (2)

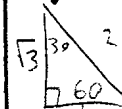
B) $(1 - \sin^2 \theta)(\tan^2 \theta + 1)$

$$= \cos^2 \theta \times \sec^2 \theta$$

$$= \frac{\cos^2 \theta}{1} \times \frac{1}{\cos^2 \theta}$$

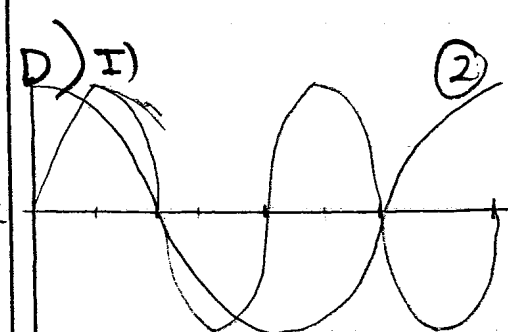
$$= 1 \quad (2)$$

(2) C) $2 \sin \theta + 1 = 0$



$$\sin \theta = -\frac{1}{2}$$

$$\theta = 210^\circ, 330^\circ \quad (2)$$



II) 5 solutions (1)

E) $F(x) = 4x^2 - 3x^2 + 6x - 1$

$$F'(x) = 12x^2 - 6x + 6$$

$$F''(x) = 24x - 6$$

$$F''(-1) = 24(-1) - 6$$

$$= -30$$

$$F'(+1) = 12(+1)^2 - 6(+1) + 6$$

$$= 12 - 6 + 6$$

$$= 12$$

$$F''(-1) - F'(+1) = -30 - 12$$

$$= -42 \quad (3)$$

Question Seven

A) $a = -5$ $d = 8$ $n = ?$

$$T_n = a + (n-1)d$$

$$283 = -5 + (n-1) \times 8$$

$$288 = (n-1) \times 8$$

$$n-1 = 36$$

$$n = 37$$

(2)

B) I) $\frac{4}{9} \div \frac{2}{3} = \frac{2}{3}$

II) $S_{\infty} = \frac{a}{1-r}$

$$= \frac{\frac{2}{3}}{1 - \frac{2}{3}}$$

$$= 2$$

(1)

C) $x+2 = x^2$

$$0 = x^2 - x - 2$$

$$= (x-2)(x+1)$$

$x = -1$ or 2

$$A = \int_{-1}^2 (x+2) - x^2 dx$$

$$= \int_{-1}^2 x + 2 - x^2 dx$$

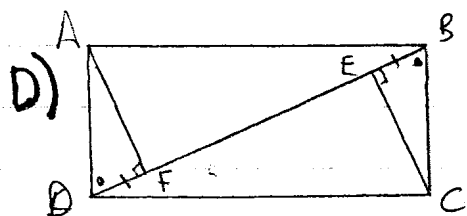
$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

$$= \left(\frac{4}{2} + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 - \frac{1}{3} \right)$$

$$= 3\frac{1}{3} - -1\frac{1}{6}$$

$$= 4\frac{1}{2} \text{ u}^2$$

(1)



I) $DF = BE$ Given

$\angle ADF = \angle BCE$ alt $\angle$'s $AD \parallel BC$

$AD \parallel BC$ opp sides of rect. are $\parallel$

$AD = BC$ opp sides of rect. are $=$

$\therefore \triangle ADF \cong \triangle BCE$ SAS (3)

II) $\angle DAF = \angle BCE$ corresp $\angle$'s of $\triangle$'s (1)

Question Eight

A) $\Delta = b^2 - 4ac = 0$

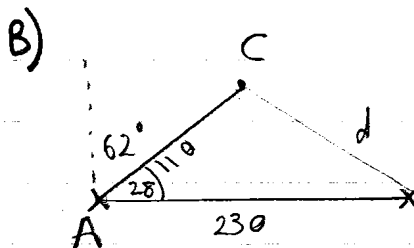
$$(k+3)^2 - 4 \cdot 1 \cdot (k+6) = 0$$

$$k^2 + 6k + 9 - 4k - 24 = 0$$

$$k^2 + 2k - 15 = 0$$

$$(k-5)(k+3) = 0$$

$$k = 5 \text{ or } -3$$



$$d^2 = c^2 + l^2 - 2cl \cos A$$

$$d^2 = 110^2 + 230^2 - 2 \times 110 \times 230 \times \cos 62^\circ$$

$$d^2 =$$

$$d =$$

(3)

C) I) $y = 2x^3 - 6x^2 - 18x + 1$

$$y' = 6x^2 - 12x - 18$$

$$y'' = 12x - 12$$

stat. point when $y' = 0$

$$0 = 6(x^2 - 2x - 3)$$

$$0 = (x-3)(x+1)$$

$x = -1$ or 3

Test using y'' or $f''(x)$

$$f'(-1) = 12(-1) - 12 \quad f''(3) = 12 \times 3 - 12$$

$$= -24 \quad = 24$$

$\therefore$ max

$\therefore$ min

Find y-values

$$f(-1) = 2(-1)^3 - 6(-1)^2 - 18(-1) + 1$$

$$= 11$$

$\therefore$ max at $(-1, 11)$

$$f(3) = 2(3)^3 - 6(3)^2 - 18(3) + 1$$

$$= -53$$

$\therefore$ min at $(3, -53)$

II) Pt of inflex when $y'' = 0$

$$0 = 12x - 12$$

$$x = 1$$

Test $x = -1 \quad 1 \quad 3$

$$y'' = -24 \quad 0 \quad 24$$

$\therefore$ change in concavity

$$f(1) = 2(1)^3 - 6(1)^2 - 18(1) + 1$$

$$= -21$$

$\therefore$ Pt of inflex at $(1, -21)$

III) increasing when $y' > 0$

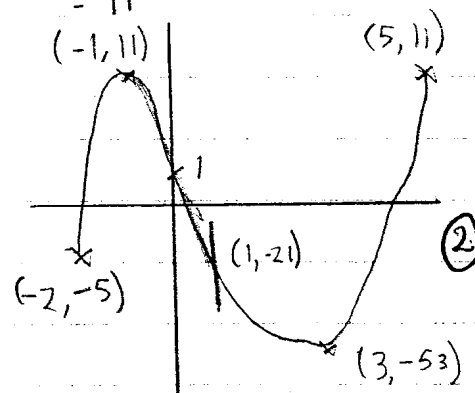
$$\therefore x < -1, \quad x > 3$$

IV) $f(-2) = 2(-2)^3 - 6(-2)^2 - 18(-2) + 1$

$$= -5$$

$$f(5) = 2(5)^3 - 6(5)^2 - 18(5) + 1$$

$$= 11$$



V) $x < 1$

(1)

Question 9

A) $\int \frac{x^3 + 6}{x^2} dx = \int x + 6x^{-2} dx$

$$= \frac{x^2}{2} - \frac{6x^{-1}}{1} + C$$

$$= \frac{x^2}{2} - \frac{6}{x} + C$$

II) $\int_3^5 (2x+1) dx = \left[x^2 + x \right]_3^5$

$$= (5^2 + 5) - (3^2 + 3)$$

$$= 30 - 12$$

$$= 18$$

Question nine

B) $\angle C = \angle ADE$ given

I) $\angle A$ is common (2)

$\therefore \triangle ABC \parallel \triangle AED$ AAA

II) $AE = 10$ pythagoras

$\frac{BC}{AB} = \frac{DE}{AE}$ corresp sides of $\parallel \Delta$'s

$\frac{BC}{15} = \frac{6}{10}$

$BC = 9$

$AC = 12$ pythag

$EC = 12 - 10$

$= 2$

C) $2x^2 - 3x + 5 =$

$A(x-1)(x-2) + B(x-1) + C$ (3)

$= Ax^2 - 3Ax + 2A + Bx - B + C$

$= Ax^2 + x(B-3A) + (2A-B+C)$

$\therefore A=2, B-3A=-3, 2A-B+C=5$

$B-6=-3, 4-3+C=5$

$B=3, C=4$

Question Ten

A) $y' = 2x - 4x^{-2}$

$y = x^2 + \frac{4}{x} + C$

$3 = 2^2 + \frac{4}{2} + C$

$3 = 6 + C$

$C = -3$

$y = x^2 + \frac{4}{x} - 3$ (3)

B) $x^2 \geq (x+1)(x+2)$

$x^2 \geq x^2 + 3x + 2$

$0 \geq 3x + 2$

$-2 \geq 3x$

$-\frac{2}{3} \geq x$

$x \leq -\frac{2}{3}$

(2)

C) $P = 2Y + 2\pi r$

$1000 = 2Y + \pi r$

$2Y = 1000 - \pi r$

$Y = 500 - \frac{\pi r}{2}$ (1)

II) $A = 2Y + \pi r^2$

$A = 2(500 - \frac{\pi r}{2}) + \pi (\frac{r}{2})^2$

$= 500\pi - \frac{\pi r^2}{2} + \frac{\pi r^2}{4}$

$= 500\pi - \frac{\pi r^2}{4}$

$\therefore = \frac{2000\pi - \pi r^2}{4}$ (2)

III) $A = 500\pi - \frac{\pi r^2}{2}$

$A' = 500 - \pi r$

$A'' = -\pi$

Let $A' = 0$ For max

$0 = 500 - \pi r$

$\pi r = 500$

$r = 500/\pi$

As $A'' < 0$ area is max

$A = 500 \times \frac{500}{\pi} - \frac{\pi}{2} \times (\frac{500}{\pi})^2$

$= 250000/\pi - 250000 \cdot \frac{\pi}{2\pi^2}$

$= 250000/2\pi \text{ m}^2$

(4)